NN
ACADEMY

IGCSE Maths Paper 4
Full Revision Paper

Cambridge IGCSE Mathematics 0580, Paper 4 Extended — calculator allowed. A complete 130-mark paper built on the weighting of the 2025 specimen and the 2025 live papers, one question per page, worked live on screen across six videos. Blank workspace sized for annotation in Xournal++.

12	130	2:30	6	16 : 9
questions	marks	exam time	videos · 2h 15m	254 × 143 mm

PREPARED FOR	DATE	SYLLABUS	NEXT SITTING
Nur Nabi	7 October 2026	0580 (2025–2027) · Extended	Paper 4 · Oct/Nov 2026

READ THIS BEFORE YOU RECORD

How to use this paper

Two ways to use it

- **Students:** sit it as a mock first — 2 hours 30 minutes, calculator allowed, all questions — then watch the videos to mark it.
- **Videos:** one video per section. Each page carries its time budget top-right: one minute per mark, a little faster than the real exam allows.

In Xournal++

- Open this PDF as the **background** (Annotate PDF), so text stays crisp at any zoom.
- Page size is **254 × 143 mm**, exactly 16:9 — a full-page view fills a 1920 × 1080 frame.
- Dark blue or black pen for working, **red only for the final answer** and the mark it earns.
- Say the method mark out loud as you earn it: "that's M1 for the cosine rule".

Everything here is original. The questions are new, written against the 0580 syllabus and modelled on the style and weighting of real Paper 4 questions. No Cambridge question is reproduced, so the videos and the worked-solutions PDF are safe to publish and monetise.

Say this in every video. This paper is a practice paper, not a prediction. No one knows what will be on the real paper — the topic mix follows the syllabus and recent papers, so every topic here is worth knowing.

Paper 4 rules (syllabus p. 59): 2 hours 30 minutes · 130 marks · **calculator allowed** · answer all questions · Extended content, including E1.14. The List of formulas is on page 2 of the real paper.

The rounding rule that costs the most marks. Give answers to **3 significant figures** unless the question says otherwise, and angles to **1 decimal place**. Never round in the middle of a calculation — keep the full value in the calculator and round once, at the end.

HOW THIS PAPER WAS BUILT

Marks by topic — the blueprint

Weighting taken from four real Paper 4s: the **2025 specimen**, **0580/41** and **0580/42 June 2025**, and **0580/42 November 2025**. "4/4" means the topic was examined in all four papers. Algebra and graphs together carry about a quarter of the marks, as they do in the real papers.

TOPIC AREA	MARKS	WHAT GETS TESTED	SEEN IN RECENT PAPERS
Number, ratio and percentages	12	Profit and loss; reverse percentage; sharing in a ratio; compound and simple interest	percentages 4/4 · ratio 4/4 · compound interest 3/4
Algebra	12	Simultaneous equations; algebraic fractions; changing the subject; factorising	rearranging 4/4 · simultaneous 3/4 · alg. fractions 3/4
Graphs of functions	12	Table of values; drawing a cubic; reading solutions off the curve; gradient by tangent	4/4 — always the longest question on the paper
Mensuration	12	Composite solid; total surface area; volume scaling; similar solids	cone/sphere 4/4 · similar solids 3/4
Trigonometry and bearings	13	Cosine rule; sine rule; bearings; area of a triangle; angle of elevation	cosine rule 4/4 · sine rule 4/4 · bearings 3/4
Statistics	12	Estimated mean; histogram with unequal widths; cumulative frequency; IQR	4/4 · histogram and CF alternate as the big one
Probability	10	Tree diagram without replacement; at least one; three selections	without replacement 4/4
Transformations and vectors	11	Describing a transformation; negative enlargement; vector geometry with ratios	vectors 4/4 · transformations 3/4
Sequences	8	Linear, quadratic, geometric and cubic nth terms in one table	3/4 — almost always a four-row table
Functions and calculus	11	Function notation; composite; inverse; exponential equation; turning points	functions 4/4 · differentiation 3/4
Travel graphs and exponential change	8	Speed–time graph: acceleration, distance, average speed; depreciation	speed–time 3/4 · depreciation 3/4
Geometry	9	Circle theorems with reasons; interior angle of a regular polygon; similar triangles	circle theorems 4/4 · polygons 2/4
Total	**130**	**12 questions, rising in difficulty inside each one**	

SYLLABUS CHECK

What this paper covers

Q	CODE	TOPIC	MK
Q01	E1.11 / E1.13	Profit, reverse percentage and ratio	6
Q01	E1.16 / E1.17	Compound interest and simple interest	6
Q02	E2.5	Simultaneous equations	3
Q02	E2.3	Algebraic fractions	3
Q02	E2.5	Changing the subject	4
Q02	E2.2	Factorising a harder quadratic	2
Q03	E2.10	Table of values and drawing a cubic	6
Q03	E2.11 / E2.12	Solving from a graph; tangent gradient	6
Q04	E5.4	Volume and surface area of a composite solid	6
Q04	E5.4 / E4.4	Volume scaling and similar solids	6
Q05	E6.5	Cosine rule and sine rule	7
Q05	E4.3 / E6.3	Bearings, triangle area, angle of elevation	6

Q	CODE	TOPIC	MK
Q06	E9.3	Estimated mean from grouped data	4
Q06	E9.7 / E9.6	Histogram, cumulative frequency, IQR	8
Q07	E8.3 / E8.4	Tree diagram without replacement	10
Q08	E7.1 / E7.2	Describing and performing transformations	6
Q08	E7.3 / E7.4	Vector geometry with a ratio	5
Q09	E2.7	Linear, quadratic, geometric and cubic sequences	8
Q10	E2.13	Function notation, composite and inverse	6
Q10	E2.12	Differentiation and turning points	5
Q11	E2.9	Speed–time graph	4
Q11	E1.17	Exponential decay	4
Q12	E4.7	Circle theorems with reasons	5
Q12	E4.6 / E4.4	Regular polygons and similar triangles	4

SYLLABUS CHECK

Not in this paper — and where to find it

> A 130-mark paper cannot test all 72 Extended subtopics, and neither can the real one. These gaps are covered by your **All of IGCSE Maths in 10 Minutes** deck and by the **Paper 2 Full Revision Paper**. Point viewers to both in the description.

CODE	TOPIC	WHERE IT IS COVERED
E1.2	Sets and Venn diagrams	10-minute deck Q02
E1.8	Standard form	10-minute deck Q04 — Paper 2 Q4 covers it
E1.9 / E1.10	Estimation, upper and lower bounds	10-minute deck Q06
E1.18	Surds	Paper 2 Q5 — excluded from Paper 4 in practice
E2.6	Inequalities and regions	10-minute deck Q15
E2.11	Asymptotes and sketching unfamiliar curves	10-minute deck Q19
E4.2 / E4.3	Constructions and scale drawing	Needs instruments — practise on paper
E4.5	Congruence and the congruence criteria	10-minute deck Q26
E6.6	3D trigonometry — angle between a line and a plane	10-minute deck Q32
E8.2	Relative frequency	10-minute deck Q35
E9.4	Box-and-whisker plots	10-minute deck Q38
E9.8	Scatter diagrams and lines of best fit	10-minute deck Q40

RECORDING PLAN

Six videos, 18–26 minutes each

VIDEO	SECTION	QUESTIONS	MARKS	RUNTIME
1	Number, ratio, percentages and algebra	Q1–Q2	24	24:50
2	Graphs of functions and mensuration	Q3–Q4	24	24:50
3	Trigonometry and statistics	Q5–Q6	25	25:50
4	Probability, transformations and vectors	Q7–Q8	21	21:50
5	Sequences, functions and calculus	Q9–Q10	19	19:50
6	Travel graphs, exponential change, geometry	Q11–Q12	17	17:50
	Whole paper	**Q1–Q12**	**130**	**2:15:00**

> Runtime = 0:30 intro + one minute per mark + 0:20 outro. Publish all six as one playlist, “IGCSE Maths Paper 4 — Full Revision Paper”, and put the blank paper PDF in every description.

CHAPTER MARKERS

YouTube chapters — paste into each description

Video 1 · Number, ratio, percentages and algebra

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q1(a)(b) Profit and reverse percentage	4:00
4:30	Q1(c) Sharing in a ratio	2:00
6:30	Q1(d) Compound interest	3:00
9:30	Q1(e) Simple interest — finding the rate	3:00
12:30	Q2(a) Simultaneous equations	3:00
15:30	Q2(b) Algebraic fractions	3:00
18:30	Q2(c) Changing the subject	4:00
22:30	Q2(d) Factorising a harder quadratic	2:00
24:30	Outro — next video	0:20

Video 2 · Graphs of functions and mensuration

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q3(a) Completing the table of values	2:00
2:30	Q3(b) Drawing the curve	4:00
6:30	Q3(c) Solving from the graph	3:00
9:30	Q3(d) Gradient by drawing a tangent	3:00
12:30	Q4(a) Volume of the composite solid	2:00
14:30	Q4(b) Total surface area	4:00
18:30	Q4(c) How many complete spheres	3:00
21:30	Q4(d) Similar solids — area to volume	3:00
24:30	Outro — next video	0:20

CHAPTER MARKERS

YouTube chapters — paste into each description

Video 3 · Trigonometry and statistics

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q5(a) Cosine rule	4:00
4:30	Q5(b) Sine rule	3:00
7:30	Q5(c) Bearings	2:00
9:30	Q5(d) Area of a triangle	2:00
11:30	Q5(e) Angle of elevation	2:00
13:30	Q6(a) Estimated mean	4:00
17:30	Q6(b) Histogram with unequal widths	4:00
21:30	Q6(c) Cumulative frequency curve	2:00
23:30	Q6(d) Interquartile range	2:00
25:30	Outro — next video	0:20

Video 4 · Probability, transformations and vectors

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q7(a) Completing the tree diagram	2:00
2:30	Q7(b) Both counters red	2:00
4:30	Q7(c) Two different colours	3:00
7:30	Q7(d) At least one blue from three	3:00
10:30	Q8(a) Describing a transformation	3:00
13:30	Q8(b) Enlargement, negative scale factor	3:00
16:30	Q8(c) Vector geometry	5:00
21:30	Outro — next video	0:20

CHAPTER MARKERS

YouTube chapters — paste into each description

Video 5 · Sequences, functions and calculus

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q9 Four sequences in one table	8:00
8:30	Q10(a)(b) Function notation and composite	3:00
11:30	Q10(c) Inverse function	2:00
13:30	Q10(d) Solving an exponential equation	1:00
14:30	Q10(e) Differentiation and turning points	5:00
19:30	Outro — next video	0:20

Video 6 · Travel graphs, exponential change, geometry

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q11(a) Speed–time graph	4:00
5:30	Q11(b) Exponential decay	4:00
8:30	Q12(a) Circle theorems with reasons	5:00
13:30	Q12(b) Interior angle of a regular polygon	2:00
15:30	Q12(c) Similar triangles	2:00
17:30	Outro — next video	0:20

List of formulas

Area, A, of triangle, base b, height h.	$A = \frac{1}{2}bh$
Area, A, of circle of radius r.	$A = \pi r^2$
Circumference, C, of circle of radius r.	$C = 2\pi r$
Curved surface area, A, of cylinder of radius r, height h.	$A = 2\pi rh$
Curved surface area, A, of cone of radius r, sloping edge l.	$A = \pi rl$
Surface area, A, of sphere of radius r.	$A = 4\pi r^2$
Volume, V, of prism, cross-sectional area A, length l.	$V = Al$
Volume, V, of pyramid, base area A, height h.	$V = \frac{1}{3}Ah$
Volume, V, of cylinder of radius r, height h.	$V = \pi r^2 h$
Volume, V, of cone of radius r, height h.	$V = \frac{1}{3}\pi r^2 h$
Volume, V, of sphere of radius r.	$V = \frac{4}{3}\pi r^3$

For the equation $ax^2 + bx + c = 0$, where $a \neq 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For the triangle shown,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\text{Area} = \frac{1}{2}ab\sin C$$

> Given in the exam — you do **not** need to memorise these. Everything else (arc length, sector area, trapezium, Pythagoras, compound interest, the sine-rule ambiguous case) must be known.

1

Number and algebra

Video 1 of 6 · Questions 1–2 · 24 marks · E1.11 – E2.5 · 24:50 target runtime

Question 1

Profit, ratio and interest

Compound interest is a multiplier raised to a power — never eight separate lines of working

(a) A shop buys a laptop for $540 and sells it at a profit of 35%. Calculate the selling price.

$ ..... [2]

(b) In a sale, the price of a printer is reduced by 18% to $287. Calculate the original price.

$ ..... [2]

(c) Amira, Bilal and Chen share $4560 in the ratio 5 : 7 : 12. Calculate the amount each of them receives.

Amira $ ..... Bilal $ ..... Chen $ ..... [2]

(d) Amira invests her share at a rate of 3.4% per year compound interest. Calculate the value of her investment at the end of 6 years. Give your answer correct to the nearest dollar.

$ ..... [3]

(e) Bilal invests his share at a rate of r% per year simple interest. At the end of 8 years his investment is worth $1968.40. Find the value of r.

$r =$ ..... [3]

Question 2

Algebraic manipulation

Every mark in this question is a method mark you can earn even with the wrong final number

(a) Solve the simultaneous equations. You must show all your working.

$$3x + 2y = 4$$
$$5x - 4y = 25$$

$x =$ ..... $y =$ ..... [3]

(b) Simplify fully. $\dfrac{3x^2 - 12}{x^2 - x - 6}$

..... [3]

(c) Make x the subject of the formula. $y = \dfrac{3x + 2}{x - 5}$

$x =$ ..... [4]

(d) Factorise fully. $6x^2 - 7x - 3$

..... [2]

2

Graphs and mensuration

Video 2 of 6 · Questions 3–4 · 24 marks · E2.10 – E5.4 · 24:50 target runtime

Question 3

Graph of a cubic

Plot every point, then join with a single smooth curve — never a ruler, never a feathered line

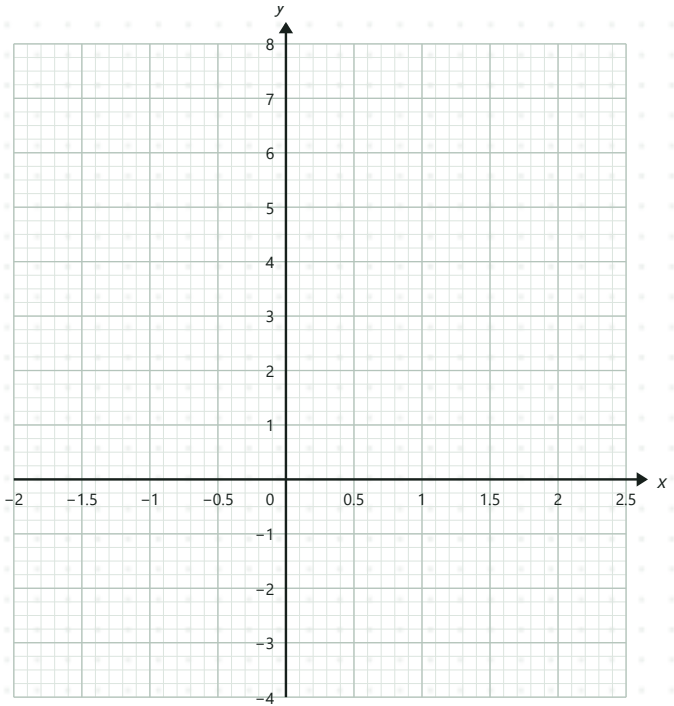
(a) Complete the table of values for $y = x^3 - 3x - 1$. [2]

x	−2	−1.5	−1	−0.5	0	0.5	1	1.5	2	2.5
y	−3		1	0.375	−1	−2.375	−3		1	7.125

Give each value correct to 3 decimal places where it is not exact.

Working space

(b) On the grid, draw the graph of $y = x^3 - 3x - 1$ for $-2 \leqslant x \leqslant 2.5$. [4]



QUESTION 3 CONTINUED

(c) Use your graph to solve the equation $x^3 - 3x - 1 = 0$.

$x =$ ..... or $x =$ ..... or $x =$ ..... [3]

(d) By drawing a suitable tangent, estimate the gradient of the graph at the point where $x = 2$.

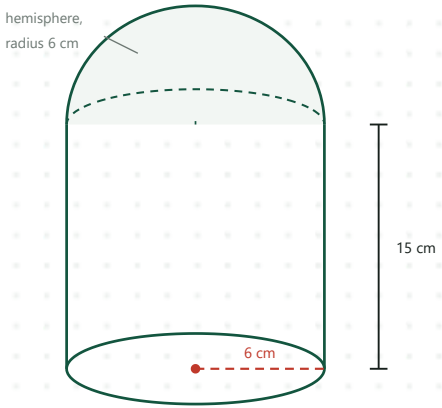
..... [3]

Question 4

A composite solid

The flat circle on top of the cylinder is covered by the hemisphere — do not count it twice

The diagram shows a solid made from a cylinder of radius 6 cm and height 15 cm with a hemisphere of radius 6 cm on top.



(a) Show that the volume of the solid is 2149 cm^3, correct to 4 significant figures.

..... [2]

(b) Calculate the total surface area of the solid.

..... cm^2 ..... [4]

QUESTION 4 CONTINUED

(c) The solid is melted down and made into spheres of radius 1.6 cm. Calculate the greatest number of complete spheres that can be made.

..... [3]

(d) A mathematically similar solid has a total surface area of 2262 cm^2. Calculate the volume of this solid.

..... cm^3 [3]

3

Trigonometry and statistics

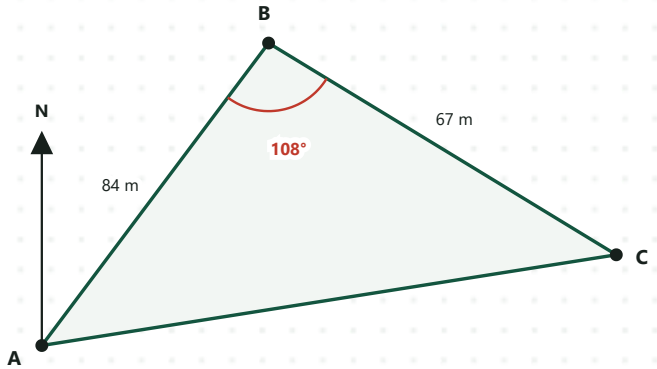
Video 3 of 6 · Questions 5–6 · 25 marks · E4.3 – E9.7 · 25:50 target runtime

Question 5

Cosine rule, sine rule and bearings

Two sides and the angle between them means the cosine rule — there is no other way in

A, B and C are three points on level ground. AB = 84 m, BC = 67 m and angle ABC = 108°.



NOT TO SCALE

(a) Calculate AC.

AC = ... m ..... [4]

(b) Calculate angle BAC.

Angle BAC = ...° ..... [3]

QUESTION 5 CONTINUED

(c) The bearing of B from A is 052°, and C lies to the east of the line AB. Find the bearing of C from A.

.....° [2]

(d) Calculate the area of triangle ABC.

..... m^2 [2]

(e) A vertical mast of height 30 m stands at C. Calculate the angle of elevation of the top of the mast from A.

.....° [2]

Question 6

Grouped data

Unequal class widths means frequency density — height is frequency ÷ width, never frequency

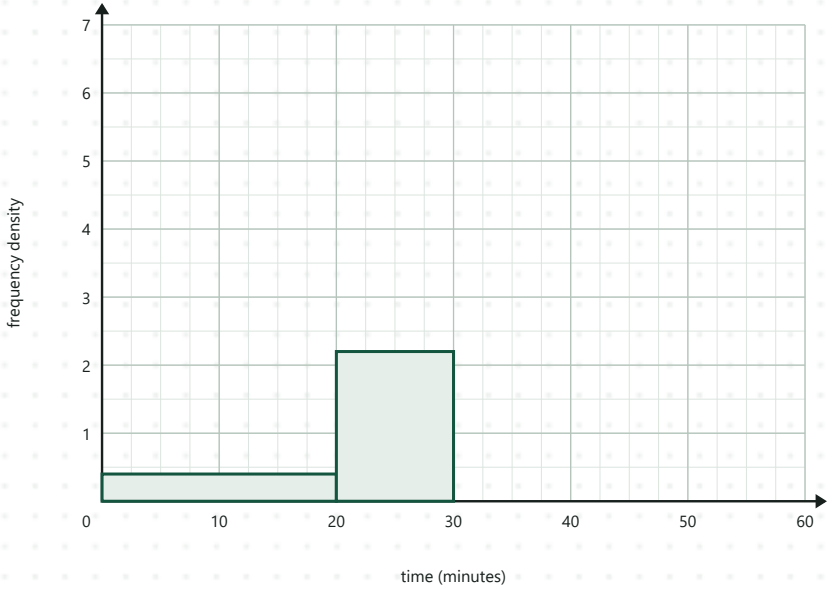
The table shows the times, t minutes, taken by 120 students to finish a test.

Time (t minutes)	$0 < t \leqslant 20$	$20 < t \leqslant 30$	$30 < t \leqslant 35$	$35 < t \leqslant 40$	$40 < t \leqslant 60$
Frequency	8	22	30	34	26

(a) Calculate an estimate of the mean time.

..... min ..... [4]

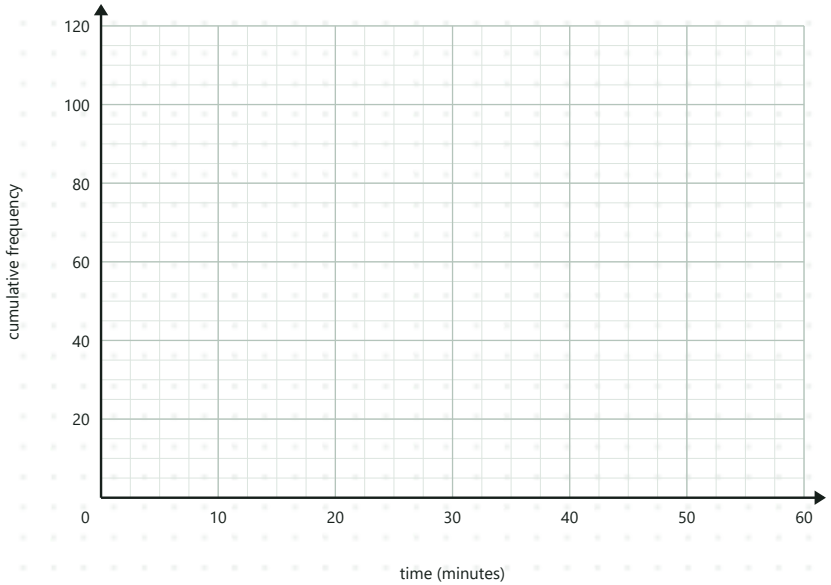
(b) Complete the histogram to show this information. The first two bars have been drawn for you. [4]



QUESTION 6 CONTINUED

(c) Complete the cumulative frequency table and draw the cumulative frequency diagram on the grid. [2]

Time (t minutes)	$t \leqslant 20$	$t \leqslant 30$	$t \leqslant 35$	$t \leqslant 40$	$t \leqslant 60$
Cumulative frequency	8				120



(d) Use your diagram to estimate the interquartile range.

..... min ..... [2]

4

Probability, transformations and vectors

Video 4 of 6 · Questions 7–8 · 21 marks · E7.1 – E8.4 · 21:50 target runtime

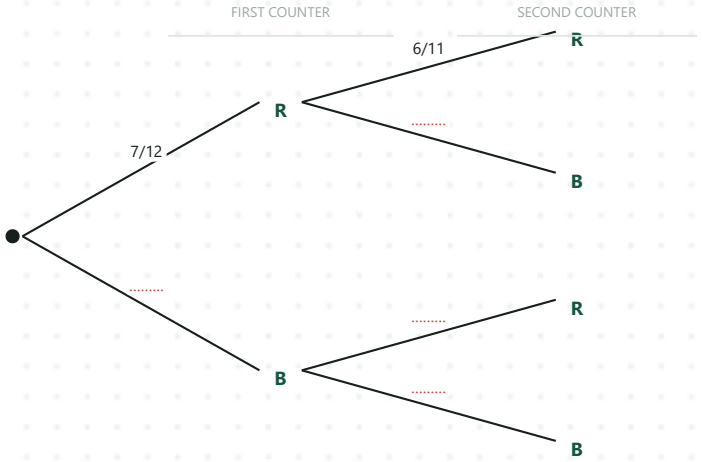
Question 7

Without replacement

"At least one" is always 1 minus the probability of none — the short way is the only way here

A bag contains 7 red counters and 5 blue counters. Two counters are taken at random, without replacement.

(a) Complete the tree diagram. [2]



(b) Find the probability that both counters are red.

..... [2]

(c) Find the probability that the two counters are different colours.

..... [3]

(d) Three counters are now taken at random from the original bag, without replacement. Find the probability that at least one is blue.

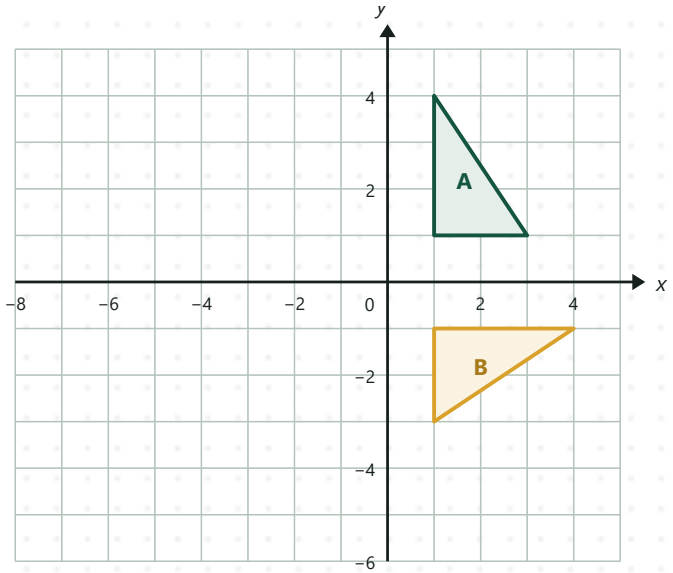
..... [3]

Question 8

Transformations and vectors

One transformation means one word plus its details — two words scores zero, however right they are

The diagram shows triangle A and triangle B on a grid.



(a) Describe fully the single transformation that maps triangle A onto triangle B.

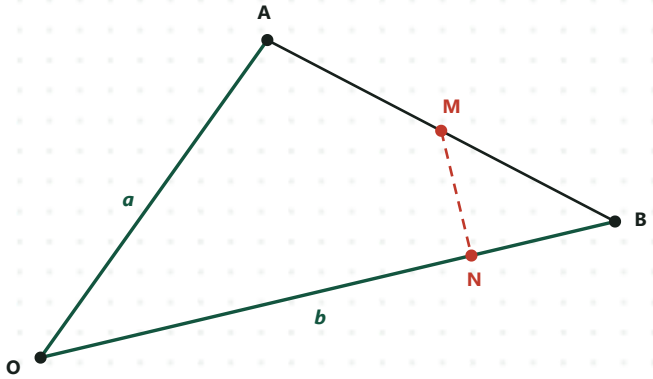
.....

[3]

(b) On the grid, draw the image of triangle A after an enlargement with scale factor −2 and centre (0, 1). Label it C. [3]

QUESTION 8 CONTINUED

In the diagram, $\mathbf{OA} = \boldsymbol{a}$ and $\mathbf{OB} = \boldsymbol{b}$. M is the midpoint of AB, and N is the point on OB such that ON : NB = 3 : 1.



NOT TO SCALE

(c) Find, in terms of $\boldsymbol{a}$ and $\boldsymbol{b}$, giving each answer in its simplest form.

(i) $\mathbf{AB}$

..... [1]

(ii) $\mathbf{OM}$

..... [2]

(iii) $\mathbf{MN}$

..... [2]

5

Sequences, functions and calculus

Video 5 of 6 · Questions 9–10 · 19 marks · E2.7 – E2.13 · 19:50 target runtime

Question 9

Four kinds of sequence

Four rows, four different rules — decide which kind it is before you write anything down

Complete the table. For each sequence, write down the next term and an expression for the nth term.

SEQUENCE	NEXT TERM	nTH TERM	
7, 11, 15, 19, ...	.....	.....	[2]
2, 6, 12, 20, 30, ...	.....	.....	[2]
3, 6, 12, 24, ...	.....	.....	[2]
1, 8, 27, 64, ...	.....	.....	[2]

Check the first differences. If they are constant the sequence is linear; if the **second** differences are constant it is quadratic; if each term is a fixed multiple of the one before it is geometric.

Working space

Question 10

Functions and turning points

fg(4) means do g first — work from the inside out, every time

$f(x) = 3x - 5$ $g(x) = x^2$ $h(x) = 2^x$

(a) Find f(−3).

..... [1]

(b) Find fg(4).

..... [2]

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... [2]

(d) Solve $h(x) = 32$.

$x =$ ..... [1]

(e) A curve has equation $y = x^3 - 6x^2 + 9x + 2$. Find the coordinates of the two turning points and determine whether each is a maximum or a minimum. You must show all your working.

..... [5]

6

Travel graphs and geometry

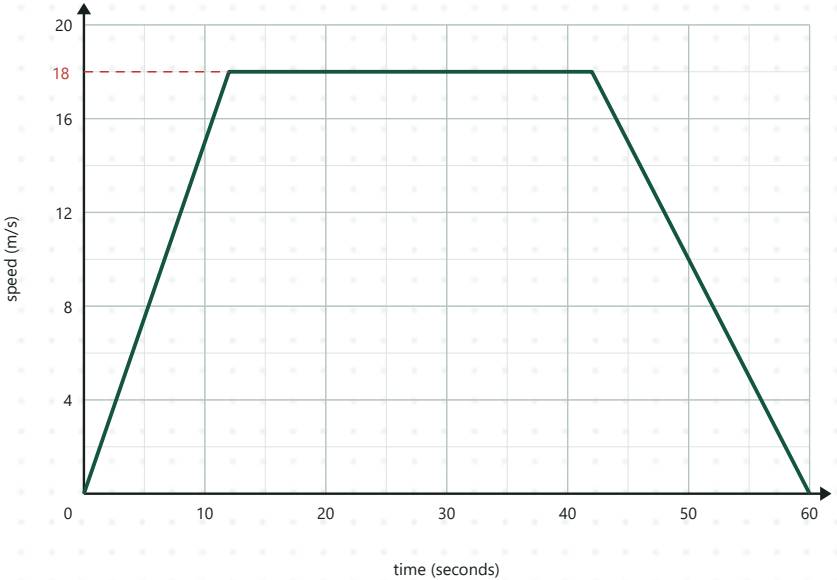
Video 6 of 6 · Questions 11–12 · 17 marks · E1.17 – E4.7 · 17:50 target runtime

Question 11

Speed–time graph and decay

Area under a speed–time graph is distance — split it into a trapezium, not three guesses

The speed–time graph shows the journey of a car.



(a) (i) Calculate the acceleration during the first 12 seconds.

..... m/s² [1]

(ii) Calculate the total distance travelled.

..... m [2]

QUESTION 11 CONTINUED

(iii) Calculate the average speed for the whole journey.

..... m/s ..... [1]

(b) A machine is worth $24 000. Its value decreases exponentially by 12% each year.

(i) Calculate its value after 5 years.

$ ..... [2]

(ii) Find the number of complete years before its value first falls below $8000.

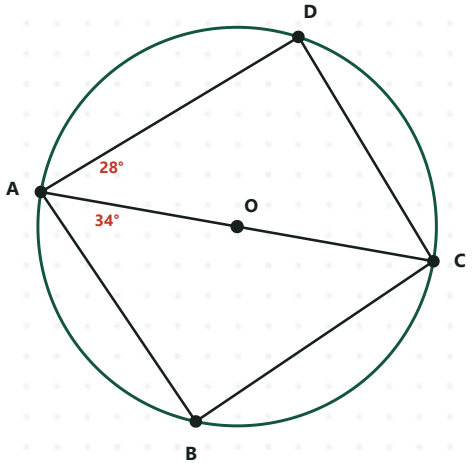
..... years ..... [2]

Question 12

Circle theorems and similarity

No reason, no mark — name the theorem in full, every single time

A, B, C and D lie on a circle, centre O. AC is a diameter, angle BAC = 34° and angle CAD = 28°.



NOT TO SCALE

(a) Find each of the following angles. Give a geometrical reason for each answer.

(i) Angle ABC

Angle ABC = ...°

Reason ..... [2]

(ii) Angle BCA

Angle BCA = ...°

Reason ..... [1]

(iii) Angle BCD

Angle BCD = ...°

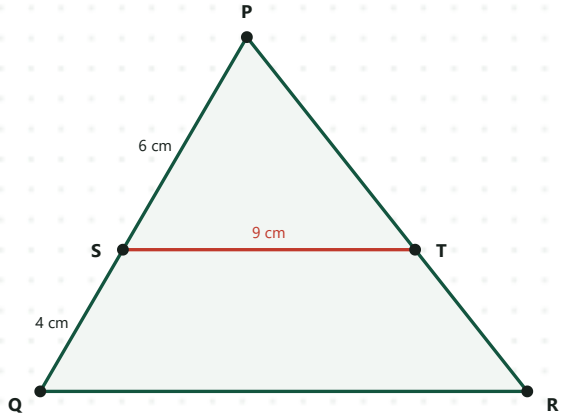
Reason ..... [2]

QUESTION 12 CONTINUED

(b) The interior angle of a regular polygon is 162°. Find the number of sides of the polygon.

..... [2]

(c) In triangle PQR, ST is parallel to QR. PS = 6 cm, SQ = 4 cm and ST = 9 cm.



NOT TO SCALE

Calculate QR.

QR = ... cm ..... [2]